

Dear Hiring Committee,

I am writing to express my strong interest in the position as “Associate Professor in theoretical or computational quantum technology” at the Department of Physics, University of Oslo (UiO).

I am a theoretical and computational condensed-matter physicist currently employed as an MSCA Postdoctoral Fellow (COFUND) at UiO, where I lead the research project DSPINN. My research focuses on many-body quantum systems, and on how unconventional quantum phases can emerge in realistic interacting systems. Quantum spin liquids (QSLs) are a class of phases I find particularly intriguing. They provide prototypical examples of fractionalisation, have shaped much of our understanding of long-range entanglement and topological order and have proposed connections to unconventional superconductivity. This combination of fundamental theoretical problems, realistic systems and the possibility of longer-term applications was what first drew me to frustrated quantum matter, and it remains the central motivation for my research.

During the last five years, I have written eleven papers, eight of them as first author and six of them since I completed my PhD two years ago. Most notably, I have demonstrated that an *exact* resonating valence bond (RVB) spin liquid emerges in the dilute doping limit of a frustrated Mott insulator [Nature Physics **21**, 1211 (2025)]. This identified kinetic frustration of dopants as a new mechanism for stabilising quantum spin liquids and has opened a research direction sitting between the quantum magnetism and correlated metals communities. Furthermore, it forms the backbone of my research programme: *Kinetic frustration as a new route to quantum matter*. The exact RVB states in the few-hole problem provides a rare, analytically controlled foundation, and the central question is what survives as one goes from this limit to finite hole density: Does the RVB physics persist? Are there fractionalisation, long-range entanglement and topological order? What competing phases appear? And can the holons, which are bosonic, survive at finite density and condense, providing a new route to superconductivity?

To address these questions, I will build a research programme at the interface of condensed-matter theory, computational physics and quantum technology. I will combine my expertise in tensor-network methods with development of classical computational methods and quantum algorithms for state preparation on near-term quantum hardware. The exact RVB states are the unifying thread throughout: they give these methods an unusually well-controlled benchmark/target before being pushed into regimes with no exact answer.

I have twice won competitive external funding based on my own project proposals: the Aker Scholarship, which funded four years of doctoral research at the University of Cambridge, and the Marie Skłodowska-Curie COFUND fellowship supporting DSPINN, the project I proposed and have led since 2025. Now, my research programme will form the foundation for applications to the Norwegian Research Council’s Young Research Talent programme, and for the ERC Starting Grant, both to be submitted in 2027.

My expertise in tensor-network methods is not yet represented at the Department, and combined with a distinctive route to quantum matter, it would complement existing activities well. My work naturally connects the Section for Condensed Matter Physics and the Section for Theoretical Physics, and I would contribute to a collective community in theoretical and computational quantum science and technology at the Department.

Experimental collaboration is something I value highly, and I wish to initiate connections with the experimental environments at the Centre for Materials Science and Nanotechnology and the Section for Solid-State Physics and Quantum Technology, including the ARPES group whose techniques directly probe the charge degrees of freedom relevant to my kinetic-frustration programme and NAFUMA who synthesise magnetic materials. I envision further synergies with UiO’s broader quantum initiatives, including QOMBINE at the Department of Mathematics, the Hylleraas Centre, and the computational community connected to dScience. Nationally, I already collaborate with researchers at the Institute for Energy Technology on neutron-scattering signatures of frustrated magnets.

Furthermore, I have a strong passion for teaching, being perhaps the most direct way I can contribute to society and make a difference. Witnessing a student understanding a problem, ideally through guided, independent reasoning, is particularly rewarding and represents the kind of learning environment I aim to foster. Beyond the impact teaching has on individual students, it also plays a central role in educating future physicists and thereby contributing to society. This perspective has shaped my interest in curriculum design and the development of coherent study programs at the institutional level. Thus, I apply not only with an aim of building an independent research programme in theoretical and computational QST at UiO, but also with a genuine wish to contribute to teaching, and towards a strong quantum education in Norway.

I have extensive teaching experience for my age, both from UiO and the University of Cambridge. Courses I have been assisting include quantum mechanics, computational physics and theoretical physics, and I am currently co-lecturing the Master's course Quantum mechanics for many-particle systems at UiO (FYS4480). I am comfortable teaching across a broad range of topics, from fundamental bachelor's courses to advanced master's courses, covering most of the relevant basic/theoretical/computational physics courses for the QST study programmes at UiO. Furthermore, my research programme includes student projects fitting well within the QST, Theoretical physics, Condensed matter physics and CS: Physics master's programmes.

I also have a passion for science communication and outreach, which is shown by, for example, my many years of engagement in the Simula Prepare Project, where I visited high schools, giving popular science talks to inspire students to choose career paths within STEM. I hope to continue participating in various activities showcasing physics to the general public, such as open days at UiO, public science talks and maintaining an active profile on social media. I also enjoy engaging with students and disseminate my research in a pedagogical and understandable way for young students.

Thank you for considering my application. I hope to get the possibility of discussing how my research programme and educational vision fits within the Department of Physics at UiO.

Kind regards,

Cecilie Glittum

## Research plan – Cecilie Glittum

### *Kinetic frustration as a new route to quantum matter*

#### 1 Research vision and core questions

The development of quantum technologies is closely linked to fundamental advances in our understanding of complex quantum states and interacting quantum matter. My research **combines analytical and numerical approaches to study quantum matter**, with a particular aim of understanding how frustration in such systems can induce exotic quantum phases such as quantum spin liquids (QSLs).

Historically, the search for QSLs has focused on frustrated magnetic exchange interactions, following Anderson's resonating valence bond (RVB) proposal. However, Hamiltonians with exact RVB ground states are often idealised and difficult to realise in materials. My recent work identified an alternative mechanism to QSL physics: kinetic frustration [4]. To minimise the kinetic energy in doped spin-1/2 Mott insulators on frustrated lattices, carrier motion generates effective antiferromagnetic correlations in the spin background [5]. I have shown that this effect **stabilises an exact RVB liquid** when doping the infinite- $U$  Hubbard model on lattices of corner-sharing tetrahedra, such as the checkerboard and pyrochlore lattices, with one or two holes (see Fig. 1). This exact result holds both for small periodic clusters as well as in the thermodynamic limit. An analytical ground state for a realistic Hamiltonian is a rare result in strongly correlated systems, and this state being an exact RVB makes the result particularly intriguing.

This realisation of doping-induced spin liquidity has opened a new avenue of research, and shapes what stands as the backbone of my research programme: **kinetic frustration as a new route to quantum matter**. The exact RVB states in the few-hole problem provide the controlled foundation, and **the central problem is what happens when we go from this limit to finite hole density**: Does the RVB physics survive? What are the effective degrees of freedom? Is there fractionalisation, an effective gauge structure, long-range entanglement or topological order? What competing phases appear? And, as one particularly striking possibility: the exact few-hole solution already indicate that holes are dressed by the RVB background to behave as effective bosonic holons. If this bosonic character survives at finite density, these holons could condense without pairing, giving a highly unconventional route to superconductivity.

Accessing these regimes is hard with conventional analytical and computational methods. Thus, development of **classical computational methods** (neural quantum states, field-theoretic methods) **and quantum algorithms/simulators** is likely essential to make progress. The exact RVB states will provide solid benchmarks for validating the classical methods before applying them to the challenging regimes, and controlled many-body targets for state preparation and benchmarking, hoping that ultimately quantum platforms may allow me to reach regimes that are difficult classically. The overall architecture of my programme is illustrated in Fig. 1.

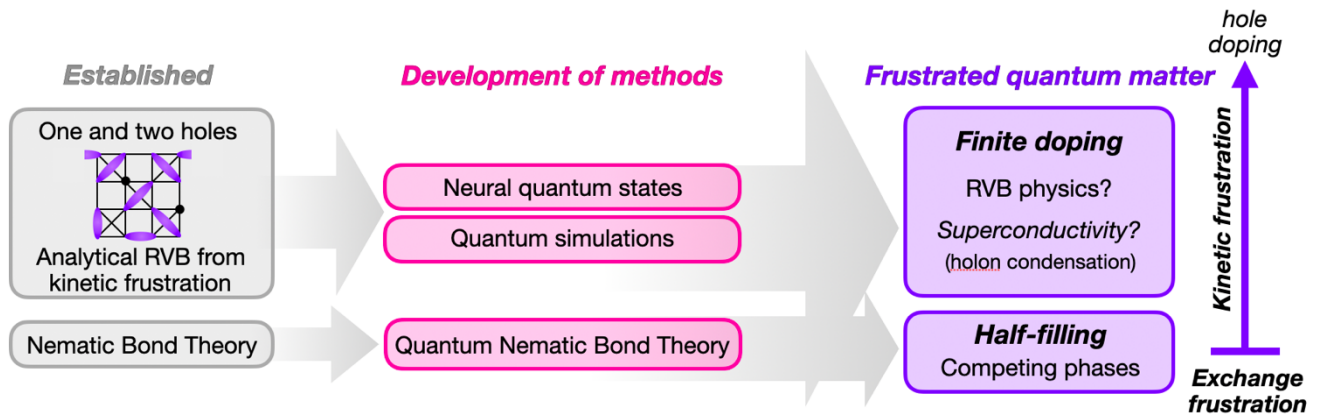


Fig. 1: The research programme builds on the realisation that *kinetic* frustration induces an exact RVB ground state on the checkerboard and pyrochlore lattices when doping the infinite- $U$  Hubbard model with one or two holes. This result will serve as benchmarks and target states for development of classical computational methods (neural quantum states) and algorithms for digital quantum simulators. These methods, together with the quantum extension of the Nematic Bond Theory, will inform the finite-doping regime as well as its competition with exchange-frustrated states near half-filling upon relaxing the infinite- $U$  constraint. The programme will establish how exotic properties of frustrated quantum matter can be realised in realistic theoretical models, and also provide signatures relevant for experimental interpretation and analogue quantum simulators.

## 2 Kinetic frustration at finite hole density

**2.1 Stability of RVB physics on cylinders:** At finite hole density, mobile carriers may preserve the RVB/QSL state, but they may also induce magnetic order, valence-bond order, charge order or phase separation. I will map the finite-doping phase diagram as a function of hole density, with the two-dimensional checkerboard lattice as the primary near-term target. With iDMRG accessing finite-density states on quasi-one-dimensional cylinder geometries, checkerboard cylinders can then readily be studied at finite density without the need of new method development. The phase will be diagnosed through spin, charge, and bond correlations to assess whether the RVB correlations dominate over the competing orders proposed above.

**2.2 Neural quantum states (NQS) for genuine two- and three-dimensional systems:** Finite doping in genuine two- and three-dimensional systems such as the checkerboard and pyrochlore lattices is beyond the reach of conventional numerical methods. To reach this regime, I will utilise NQS [8][9] to parameterise variational wave functions, whose parameters are optimised using variational Monte Carlo. Such methods have outperformed state-of-the-art tensor-network methods by several orders of magnitudes in the number of parameters used for frustrated spin systems [10][11][12][13][14]. Recently, NQS were, for the first time, also applied to simulate the  $t$ - $J$  model on the two-dimensional square lattice [15], giving similar indications. Notably, the entire doping regime was accessed. Although the applicability of NQS to the  $t$ - $J$  model remains broadly unexplored, all these recent advancements suggest that NQS could potentially be used also for the checkerboard and pyrochlore  $t$ - $J$  models to explore larger system sizes and access the intermediate doping range. The applicability of such methods to new systems is generally difficult to predict [16]. However, the exact doped RVB states, available at all system sizes, give this effort an unusual advantage: they are controlled benchmarks with known energy, amplitudes, correlation functions, meaning that the various NQS ansatzes can be verified in this limit before applied at finite doping.

For fermionic models, determinant-based ansatzes such as Hidden Fermion Determinant States [17], which was used for the  $t$ - $J$  model in Ref. [15], Jastrow-factor-based methods [18][19] and neural backflow transformations [20] provide a natural way to incorporate fermionic statistics. Validated ansatzes will first be applied to larger systems and hole densities for the checkerboard lattice, and later the pyrochlore lattice, where a key goal is to reach large enough sizes to recover the lattice loop structure that is strongly distorted in small periodic clusters.

**2.3 Emergent properties of doping-induced RVB liquids:** The exact few-hole state exhibits spin-charge separation. I will further study whether its local dimer constraints support an effective gauge-theory description with long-range entanglement and topological order, using signatures accessible to iDMRG: the symmetry-resolved entanglement spectrum and the topological entanglement entropy (extracted via Rényi entropies on suitable bipartitions), together with flux-insertion on cylinders to test for topological ground-state degeneracy. This diagnostic toolkit is native to iDMRG and will be applied first to checkerboard cylinders. In two and three dimensions, I will rely on NQS for ground-state energies, correlation functions and static structure factors. The extension of entropy and flux-response diagnostics to NQS are open questions I will pursue as the field matures.

**2.4 A highly unconventional route to superconductivity:** I will derive effective models for the dressed holes and compare their predictions with numerical spectra and correlation functions, such as hole-hole correlations, to determine their effective statistics and interactions. If bosonic holon degrees of freedom persist also at finite density, their condensation could provide a highly unconventional route to superconductivity, where holons condense without hole-pairing. I will test this directly through the long-distance correlations of singlet-pair operators, resolved by bond orientation to determine electron-pairing symmetry. On iDMRG cylinders, superconducting tendencies will be identified when electron-pair correlations decay algebraically and more slowly than competing spin, charge, and valence-bond correlations and NQS calculations will test persistence in two- and three- dimension. A positive result here would identify a genuinely new mechanism for superconductivity. **ADD?**

**2.5 Quantum algorithms with a controlled target:** The computational complexity of quantum many-body systems severely limits the applicability of classical numerical methods. This holds particularly for frustrated systems finite doping, larger two- and three-dimensional systems and for computations of real-time dynamics. Quantum computers may ultimately provide a complementary route beyond what classical methods can reliably simulate. Although quantum advantage remains a longer-term goal, the exact few-hole RVB states provide a well-controlled and physically relevant target for developing and benchmarking quantum algorithms, which also

provides an opportunity to quantify circuit depth and gate requirements, and build expertise for the future use of quantum hardware in many-body computations.

The current central challenge in quantum computing is reliable state preparation. Since the doped RVB state is analytically known for small system sizes, and can be written as an exact matrix product state (MPS), it provides an unusually useful starting point for benchmarking such algorithms. Matrix product states (MPS) enable efficient quantum state preparation, allowing complex states to be encoded into quantum circuits with dramatically reduced depth and gate counts [24]. I will first benchmark MPS-based preparation. However, as preparation of states with more holes will require more general approaches, I will further benchmark variational and adiabatic protocols. One general approach I am particularly interested in pursuing is to adapt resolution-refinement [28] to frustrated lattices with a multi-site unit cell, using the result as input to the Rodeo algorithm [29]. Rodeo then filters a target energy interval through controlled time evolution and ancilla measurements. Its performance depends on the initial overlap with the target eigenstate, which is why a resolution refined initial state will make it perform better (*manuscript in preparation*).

Once a target state can be reliably prepared, I will evaluate measurement protocols for dynamical observables, first focusing on the dynamical spin structure factor, obtained by measuring unequal-time spin correlations and Fourier transforming, providing signatures that distinguish doping-induced RVBs from exchange-frustrated QSLs. Subsequently, I will focus on the single-particle spectral function, exploiting recent proposals for measuring ARPES spectral functions directly on quantum hardware, giving direct access to the holon dispersion [30]. These measurements will test whether the low-energy spin and charge dynamics of doping-induced RVB states can be resolved on quantum hardware.

### 3 Competing phases at finite $J$

The exact RVB liquid arises in the infinite- $U$  limit, where the system is governed by the kinetic energy of the dopants/holes. Realistic Mott insulators will have finite  $U$ , generating a Heisenberg-exchange term if  $U$  is sufficiently large, i.e. a  $t$ - $J$  model. In the case of unfrustrated hopping, kinetic energy favours (Nagaoka) ferromagnetism, and with antiferromagnetic  $J$ , the competition between phases is very clear. In the case of kinetic frustration however, both phenomena favour antiferromagnetism, and therefore the competition is less, giving hope for stronger stability of the RVB wave function. Indeed, as shown in Ref. [4], the doping-induced RVB for the one- and two-hole case is stable upon inclusion of small antiferromagnetic nearest-neighbour exchange  $J$ . Kinetic magnetism and exchange are however likely to favour different kinds or (frustrated) antiferromagnetic states on the lattices of interest, and will compete. As  $U$  increases, frustration from exchange becomes more important, leading to a direct competition between the two types of frustration. Realistically, relaxing the infinite- $U$  constraint therefore takes the kinetic-frustration programme towards a regime where exchange-frustrated states become more important and, at half filling, to the frustrated spin-1/2 Heisenberg model. I want to investigate and understand this competition, starting with the frustrated spin-1/2 Heisenberg model.

**3.1 Long-standing questions in exchange-frustrated quantum magnets:** The spin-1/2 checkerboard and pyrochlore Heisenberg antiferromagnet are paradigmatic examples of exchange-frustrated models with extensively degenerate classical counterparts, however, particularly for the pyrochlore case, quantum ground states, low-energy excitations, and finite-temperature correlations are still being debated. Together, the remaining open questions regarding these paradigmatic models motivates the development of new computational methods tailored to exchange-frustrated systems. NQS form the principal near-term direction also here.

**3.2 Quantum Nematic Bond Theory (qNBT):** For the past eight years, I have contributed to the development of Nematic Bond Theory (NBT), a self-consistent diagrammatic field theory for *classical* frustrated magnets [21]. The method has proven powerful for frustrated classical systems, including the discovery of new phases and phase transitions in pyrochlore [23] and triangular [24][25] Heisenberg magnets. Its prediction of a spiral spin liquid on the triangular lattice [24] has recently gained experimental support from neutron-scattering on  $\text{AgCrSe}_2$  [26]. A different route to spin-1/2 Heisenberg models would be to construct a quantum extension of NBT, qNBT.

NBT goes beyond the Self-Consistent Gaussian approximation [22] by treating the local unit-spin-length constraint through a fluctuating constraint field, allowing for spontaneous point-group-symmetry breaking. This treatment leads to a set of self-consistent equations, which can be efficiently solved numerically to give the free energy and static structure factor versus temperature. This enables for unusual direct comparison of the free energy of

competing phases. I will extend this approach to quantum spins by combining NBT with a Schwinger-boson formulation. The aim is to establish a practical framework for computing thermodynamic and dynamical properties of frustrated quantum magnets in two and three dimensions. qNBT will contain Hubbard-Stratonovich bond-fields in addition to constraint-fields fixing the local boson occupation number. Going decisively beyond standard Schwinger-boson mean-field theory, the fluctuations of the bond-field will be treated together with the constraint field in the style of NBT to give a set of self-consistent equations to be solved numerically. It will be benchmarked initially against established numerical approaches on accessible models, such as the unfrustrated spin-1/2 square antiferromagnet, before being applied to parameter regimes beyond their reach.

This is the most ambitious and highest-risk component of the research programme. However, its potential payoff is substantial, as qNBT could provide a new route to thermodynamic and spectral observables in frustrated quantum magnets, potentially answering long-standing questions in exchange-frustrated systems and also directly supporting comparisons with experiments (see Section 4). In particular, I will compute static and dynamical spin structure factors and thermodynamic responses. In developing qNBT, central focus will be on the free energy (to access the heat capacity *and to be able to directly compare the free energy of competing states*), the static structure factor and the dynamical structure factor as functions of temperature.

Once the competing phases are established, the natural question is how the phases identified at finite doping evolve into the exchange-dominated phases as  $J$  increase. Such a study could be attempted with NQS. qNBT/NQS could also serve as cross-checks of each other: if they agree, the two approaches are likely to have found the correct physics, if not, the NQS ground state could be attempted as an initial bias for qNBT to compare free energies.

## 4 Material realisations and quantum simulators

**4.1 Material realisations:** A major obstacle in experimental detection of QSLs is the unambiguous identification of QSL signatures in experimental data. As a theorist, I can contribute by computing observables directly comparable with experiment to characterise signatures of spin-liquid correlations as well as nearby competing phases. As already mentioned throughout the plan, I will compute static and dynamical spin structure factors (for neutron scattering experiments), thermodynamic responses and quantities such as hole density, holon dispersion and hole-hole structure factors (for ARPES/RIXS/STM). *(Bond correlators in quantum gas microscopy?)*

A further objective is to guide experimentalists in the search for candidate materials. Thus, the robustness of theoretically identified QSLs should be tested against material-relevant perturbations, including exchange anisotropy, longer-range interactions, lattice distortions, and spin-orbit coupling. The aim is to identify which perturbations preserve, select, or destroy the QSL physics and to formulate experimentally distinguishable consequences. This is particularly relevant because candidate spin-1/2 pyrochlores can display substantial anisotropy and spin-orbit-driven interactions rather than an ideal isotropic Heisenberg Hamiltonian.

**4.2 Analogue simulators:** Analogue quantum simulators provide intriguing platforms for simulating specific many-body Hamiltonians. Also here, the doping-induced RVB provides an uncommon ability to benchmark simulator experiments to then access larger systems and the finite-doping regime. Although fermionic simulators are becoming accessible, bosonic systems are easier to cool down, and also accessible in Rydberg arrays. It would therefore be interesting which properties of the fermionic ground state carries over to the bosonic counterpart. In the single-hole case, bosonic statistics modify the relative sign of different dimer configurations, while retaining the RVB ground state (through a generalised Jordan-Wigner transformation, *manuscript in preparation*). I will determine how this modification extends to two and more holes, and whether kinetic frustration can stabilise QSL phases in experimentally accessible regimes also for spin-1/2 bosons. The checkerboard lattice can in principle be made with cold atoms and Rydberg arrays. Furthermore, I am in contact with an experimental group at the University of Cambridge currently studying triangular-based bosonic cold atomic systems, such as the kagome lattice, so to investigate kinetic frustration on such a system would also be of direct experimental relevance.

## 5 Possible synergies

My research programme connects the Section for Theoretical Physics and the Section for Condensed Matter Physics, and I wish to deepen the interactions between sections, contributing to a collective environment in theoretical and computational QST at the Department. I bring tensor-network expertise not yet represented locally, together with a distinctive route into QSLs and quantum matter grounded in an exact, analytically controlled result.

My tensor-network and strongly-correlated-systems expertise complements Prof. Morten Hjorth-Jensen and the EIGEN-Q environment's work on quantum computing algorithms, quantum many-body theory, machine learning and NQS, particularly relevant to Section 2. Further, I complement Prof. Olav F. Syljuåsen's expertise in exchange-frustrated magnetism and quantum Monte Carlo (QMC), enabling cross-validation between qNBT, NQS, DMRG and QMC. He will also be a central collaborator in the development of qNBT.

The experimental community within the Centre for Materials Science and Nanotechnology (SMN) at UiO hosts several experts with whom I envision synergies. Prof. Martin Valldor and Dr. Jan Schultheiß synthesise and characterise magnetic materials, and they have interest in materials relevant to the programme. Furthermore, SMN host experts on ARPES (Dr. Simon Cooil and Prof. Justin Wells), a technique relevant for studying the charge degrees of freedom in kinetic systems. The go-to experimental method for studying magnetic properties is neutron scattering. Although there is no neutron-scattering activity at UiO, there is expertise at the Institute for Energy Technology (IFE), an institution with close ties to UiO. There, Dr. Øystein Fjellvåg does neutron scattering experiments on spin liquid candidates. He was involved in the experimental detection of spiral spin liquids [26] and we are currently collaborating on  $\text{CeBr}_3$ , a frustrated triangular layer magnet with gapless spinons. With the opening of the scientific program at the European Spallation Source in Lund, Sweden in 2027, there will be increasing experimental activity on magnetism in Scandinavia, with spectrometers accessing ultra-low energies [31], holding great potential for resolution of QSL signatures such as fractionalised excitations.

The QOMBINE project at the Department of Mathematics and the Hylleraas centre at the Department of Chemistry offers a further link to researchers in mathematics, physics, and chemistry working on quantum computation and many-body methods. Furthermore, the dScience community will foster interactions between research fields at the Faculty working with numerical methods, and in particular machine learning approaches.

The programme will also involve UiO students through MSc projects relevant for four different MSc programmes (Quantum Science and Technology, Theoretical Physics, Condensed Matter and Computational Science), spanning analytical theory, numerical many-body physics, machine learning and quantum simulations.

Lastly, the programme will benefit from collaborations with my international network of both theorists and experimentalists including researchers at the University of Cambridge, the University of Tennessee, Helmholtz Zentrum Berlin, the University of Tokyo, the University of Warwick and the University of Oxford [4][32][33].

## 6 Implementation

**6.1 Research leadership and people:** The programme cover work for several years for more than one researcher. I will lead all components while keeping an active role in the research itself. The programme divides into projects at the right scales for a group: benchmarking NQS, iDMRG on cylinders, and classical phase diagrams for exchange-frustrated models, are well-defined MSc projects; the finite-doping phase diagram and the excitation structure of doped RVB form a coherent PhD project, as does the qNBT development. The quantum algorithms, which requires work across many-body theory and quantum algorithms, is likely better suited for a postdoc. Once I have built a group, we will meet weekly alongside individual supervision, and I will give particular attention to students who must finish within a fixed period. Each project will be constructed with mitigations identified in advance so that a defensible result is reachable even if the most ambitious objective is not.

**6.2 Funding:** To build the group I will actively apply for external funding, particularly to the Research Council of Norway's Young Research Talents scheme, with submission planned for early 2027, and for an ERC Starting Grant in October 2027. The applications will build on the RVB extension to finite doping at infinite- $U$ . The programme is designed to deliver publishable results and MSc projects independently of the outcome of these applications, while external funding will enable its expansion through PhD students and postdoctoral researchers.

**6.3 Computational resources:** I will apply to SIGMA2 and the National Research Infrastructure Services for CPU, GPU and QPU allocations on the Saga, Olivia and LUMI-IQ supercomputers. I am also co-applicant in EIGEN-Q's proposal for research access to QuNorth's Magne, the next Nordic quantum computer.

**6.4 Timeline:** Years 1–2 will apply iDMRG for finite doping, benchmark NQS for the few-hole RVB states, extend validated NQS methods to finite doping and benchmark quantum state preparation. Years 3–5 will develop qNBT in parallel with further exploration of kinetic frustration and quantum algorithms for dynamical properties.

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## Summary of three selected scientific publications – Cecilie Glittum

### 1. A resonant valence bond spin liquid in the dilute limit of doped frustrated Mott insulators

Nature Physics **21** (8), 1211-1216 (2025), C. Glittum, A. Štrkalj, D. Prabhakaran, P. A. Goddard, C. D. Batista and C. Castelnovo

The ideas of the resonating valence bond (RVB) spin liquid and spin-charge separation were first put forward by Anderson and collaborators about 50 years ago. These ideas were highly transformative and impacted a broad range of areas of research throughout the years, from superconductivity to topological phases and quantum computing. Yet, while generally believed true, the existence of an RVB liquid phase in a realistic system was never hitherto demonstrated.

In this work, we show that the RVB spin liquid emerges naturally in the simple context of a doped Mott insulator, in presence of frustrated hole hopping; a context of direct experimental relevance for instance in pyrochlore materials. We provide an analytical proof in the thermodynamic limit as well as numerical evidence (employing ED and DMRG) for finite size systems. Remarkably, we are able to propose a ground state wave function in the low doping limit that appears to be numerically exact also in finite size systems. Analytical solutions are rare for strongly correlated systems, holding promise for further theory-experiment research progress. In particular, the exact solutions also in finite size systems provide solid benchmarks and target states for development of new classical computational methods as well as analogue and digital quantum simulator implementations.

While much attention has been devoted to the emergence of new phases of matter from geometrically frustrated exchange interactions, our work demonstrates that *kinetic energy frustration alone* in doped Mott insulators may be pivotal to stabilising RVB physics, opening a largely new and unexplored avenue of research. This is largely uncharted territory because it sits between the quantum magnetism and correlated metals communities, and it promises to seed collaborations between the two communities. The idea of robust topological phases induced by frustrated kinetics rather than exchange interactions has the potential to produce transformative impact in many different areas of research and technology. For example, if persisting to finite hole density, it may give rise to a highly unconventional form of superconductivity, where holons condense in absence of hole-pairing.

Since published, this work has received a lot of interest. For instance, Google Quantum AI has contacted us and expressed interest in preparing the RVB states and computing their dynamics on their quantum computer, and an experimental group in Canada has initiated studies to look for signatures of the RVB state in pyrochlore materials.

I was awarded the American Physical Society Outstanding Dissertation in Magnetism Award 2025 for this work.

### 2. Arc-shaped structure factor in the J1-J2-J3 classical Heisenberg model on the triangular lattice

Physical Review B **104**, 184427 (2021), C. Glittum and O. F. Syljuåsen

This work made two connected contributions to the study of frustrated magnetism: it established Nematic Bond Theory (NBT) as an efficient thermodynamic framework for highly degenerate classical spin systems, and it predicted unconventional correlation regimes in the triangular-lattice  $J_1$ - $J_2$ - $J_3$  Heisenberg model.

Methodologically, we extended NBT to calculate the thermodynamic free energy as a function of temperature, which is rarely directly accessible in numerical treatments of frustrated magnets. This enables unambiguous distinguishment of first-order from continuous phase transitions and determination of latent heats and other thermodynamic quantities such as the specific heat from a single framework. This makes NBT a practical tool for quantitatively characterizing phase transitions and scanning model parameter spaces in frustrated magnets at considerably lower computational cost than large-scale Monte Carlo simulations.

Physically, the paper identified a parameter regime where the energy minima of the Fourier-transformed exchange coupling form a ring of spiral wave vectors. Here, NBT predicted for the first time a spiral spin liquid on the triangular lattice. In this regime, short-range spiral correlations produce a ring of diffuse intensity in the static spin structure factor, giving a direct and distinctive neutron-scattering signature. On cooling, we further predicted an intermediate arc regime in which the scattering ring develops a broad, symmetry-breaking arc before evolving into conventional single- $q$  spiral order. This demonstrated that the lifting of spiral degeneracy need not occur through a direct transition

from an isotropic liquid to a fully ordered state, but can proceed through a partially ordered regime with pronounced orientational correlations.

The significance of these predictions has been strengthened by subsequent experiment. Single-crystal neutron-scattering measurements on the triangular-lattice material  $\text{AgCrSe}_2$  reported a continuous ring of magnetic scattering characteristic of a spiral spin liquid and identified the material as a close realisation of the frustrated triangular-lattice  $J_1$ - $J_2$ - $J_3$  Heisenberg model. The experimental work explicitly cites our 2021 paper as a theoretical prediction for the triangular-lattice spiral-spin-liquid scenario. Although the experimentally observed low-temperature evolution is described as a gradual crossover rather than the sharp phase-transition sequence predicted for the idealized model, the observation provides important material-level support for the central prediction: frustration and near-continuous spiral degeneracy can stabilize liquid-like spiral correlations with directly measurable momentum-space signatures.

The extension of the NBT method has further proven powerful for frustrated classical systems, including the discovery of new phases and phase transitions in Heisenberg magnets on several lattices [11], where in particular the access to the free energy enables distinction between stable and metastable states, particularly useful in frustrated systems, where large number of states lie close in energy.

### 3. Fractional vortices in a spin-isotropic spiral spin liquid

[arXiv:2609.XXXX](#) (2026) [submitted to Nature Communications], **C. Glittum**, H. Yan and J. Reuther

This work identifies a new class of topological excitations in spin-isotropic spiral spin liquids. While spiral spin liquids are characterized by a continuous manifold of low-energy spiral wave vectors and have been observed through ring-like magnetic-scattering signatures in several frustrated magnets, their intrinsic real-space defect structure has remained poorly understood. We show that the coupled momentum and spin degrees of freedom of a Heisenberg spiral spin liquid give rise to a nontrivial order-parameter manifold,  $[S_{\mathbf{q}}^1 \times SO(3)]/\mathbb{Z}_2$ , in which reversal of the local spiral wave vector is tied to reversal of the spiral-plane normal.

This structure allows elementary **fractional momentum vortices** carrying a  $\pi$ , rather than  $2\pi$ , winding of the local wave-vector orientation. Each fractional vortex also has a binary chirality associated with the rotation of the local spin frame. The full defect spectrum is classified by  $\mathbb{Z} \times \mathbb{Z}_2$ , with conventional momentum vortices and  $\mathbb{Z}_2$  spin vortices appearing as composites of the elementary fractional defects. Opposite fractional vortices can fuse either into the trivial sector or into a  $\mathbb{Z}_2$  vortex, producing fusion rules formally analogous to those of Ising anyons. This is a classical topological analogue rather than a realization of quantum non-Abelian topological order, but it demonstrates how anyon-like fusion structures can emerge from the topology of a classical frustrated magnet.

Our simulations establish that the fractional vortices are thermally populated throughout the correlated spiral-spin-liquid regime and remain as tightly bound pairs even after lattice-nematic order develops at lower temperature. Because the underlying mechanism requires only a near-circular spiral manifold and approximate Heisenberg spin symmetry, it should apply broadly to frustrated two-dimensional magnets, including triangular- and honeycomb-lattice spiral-spin-liquid candidates. The results therefore provide a new framework for interpreting the finite-temperature textures, dynamics and eventual ordering of spiral spin liquids, and motivate direct searches for fractional vortices using real-space magnetic imaging.

### References:

- Andriushin et al., Nat. Commun. **16**, 2619 (2025).
- C. Glittum** and O.F. Syljuåsen, PRB **108**, 014413 (2023).
- C. Glittum** and O.F. Syljuåsen, J. Phys.: Condens. Matter **38**, 135801 (2026).
- C. Glittum** and O.F. Syljuåsen, arXiv:2605.26779 [cond-mat.strl-el] (2026).
- Y. Liu, X.-P. Yao and Gang Chen, PRB **106**, L220410 (2022).

# Educational portfolio – Cecilie Glittum

## 1. Educational competence and experience

### Teaching experience

- **FYS 4480 – Quantum Mechanics for Many-Particle Systems [University of Oslo (UiO)] Autumn 2026 (38 h).**  
*Co-responsible for lecturing, group sessions and organising the course.*  
Master level course recommended for both theoretical and computational Quantum Science and Technology (QST) students covering a range of topics from diagrammatic approaches and mean field theory to DMRG and quantum computing algorithms. Responsibility on lecturing DMRG (8h of lectures), constructing problem sets and exams, in addition to attending all group sessions.
- **FYS 4110 – Advanced Quantum Mechanics (UiO) Autumn 2025 (2 h).**  
2h lecture on Quantum magnetism and magnons.
- **Part II Computational Physics [University of Cambridge (UniCam)] Spring 2023 (4 h).**  
Problem solving classes in 25% of the course: neural networks. Digital supervision for the entire cohort (~100 students).
- **Theoretical Physics I (UniCam) Autumn 2020, 2021, 2022, 2023 (8 h each year).**  
Problem solving classes for the full cohort (~100 students).
- **Advanced Quantum Physics (UniCam) Autumn 2020 (10 h).**  
Small-group supervisions with three students.
- **Writing Science – Academic writing for physics MSc students (UiO) Autumn 2019 (13 h).**  
Led sessions on academic writing, reading strategies and transition to the master's project work. Two full days of training, in addition to having attended the course myself in 2018.
- **FYS-MEK 1110 – Mechanics (UiO) Spring 2017 (40 h) and Spring 2019 (30 h).**  
*Completed the MN Learning Assistant course (~15 h) with focus on active learning, group dynamics and student engagement.*  
Learning assistant. Grading of midterm and final examinations.
- **FYS2140 – Quantum Physics (UiO) Spring 2018 (2h).**  
2h lecture summarising the course and discussing previous exam questions.
- **MAT-INF 1100 – Modelling and Computations (UiO) Autumn 2017 (60 h).**  
Teaching assistant.

### Supervision experience

- Co-supervisor of MSc student at UiO (2025-2027)  
Theme: Kinetic frustration on kagome lattices.
- Supervising a 3-month summer project for a second-year BSc student at UiO (2026)  
Theme: Numerical study of spin dipole-interactions on the pyrochlore lattice
- Main supervisor of MSc student from École normale supérieure de Lyon Spring 2025 while in Berlin  
Title: Study of vortices in XY magnets on the triangular lattice via classical Monte Carlo simulations

### Teaching development, programme and quality work

- **Designed the problem set on neural networks for Part II Computational Physics, UniCam 2023**
- **Project on multiple-choice questions in physics examinations (UiO) 2020**  
Used statistical methods to compare student performance on multiple-choice and long-answer questions in three physics examinations. Found moderate to high correlation (0.6–0.7) between formats and identified which types of multiple-choice questions contributed most positively. Co-authored a report shared with the Faculty leadership and presented at the Physics Department seminar in October 2020.
- **NOKUT evaluation of labour-market relevance in disciplinary programmes (BSc in Physics and Astronomy, UiO) Student representative 2019-2020**  
Participated in two full-day seminars organised by NOKUT, took part in the local evaluation and co-authored the report submitted by the Department of Physics, UiO.
- **Working group on the MSc programme in Physics (UiO) 2019–2020**

Evaluated the effect of the reduction in the number of master's specialisations, the Writing Science course and co-location of students.

- **Glittum-Haugerud report on the Physics and Astronomy BSc programme (UiO) 2019**  
Co-authored a report proposing changes to the structure and curriculum of the bachelor's programme in Physics and Astronomy. Focused on improving student learning, participation and labour-market relevance. Presented the report at the Physics Department seminar in October 2019.
- **Development of problem set for IN1900 – Introduction to Programming with Scientific Applications and IN1910 – Programming with Scientific Applications (Simula/UiO) 2018**  
One of two who designed a full new problem set for IN1900 and parts of a problem set for IN1910. The IN1900 problem set is still in use, and all students in the Physics and Astronomy and Quantum Technology BSc programmes meet this problem set their first semester ([link](#)).

### Committees and boards

- **Faculty Board, Faculty of Mathematics and Natural Sciences (UiO), Fixed-term staff rep. 2026**
- **Liaison Committee, Trinity College (UniCam) PhD student rep. and secretary 2021-2022**  
Official college body connecting student members with management
- **BA Committee, Trinity College (UniCam) Secretary 2021-2022**  
Arranging events for ~400 graduate students, including organisation of welcome week
- **Programme Council for the BSc in Physics and Astronomy (UiO) 2019-2020**  
Discussions on programme structure and following up the Glittum-Haugerud and NOKUT reports
- **Programme Council for the MSc in Physics (UiO) 2018-2019**
- **Department Board, Department of Physics (UiO), Deputy student representative 2017-2020**

### Outreach, recruitment and public engagement

- **Outreach activities at UiO**  
Welcome-week seminar for bachelor's and master's students in Quantum Technology August 2026  
Fysikk i Origo (seminar series organised by the physics student associations) May 2026  
Seminar on academic career for final-year master's students January 2026  
Seminar on academic career for new master's students in physics August 2025
- **Simula Prepare Project 2016-2019**  
Student ambassador visiting lower and upper secondary schools in the Oslo region to give talks on natural sciences and inspire towards STEM studies

## 2. Reflection note on teaching practice and view of learning (*anchored in SoTL*)

By the time physics students leave university, they should be able to carry out familiar calculations, but more importantly, they should be able to approach problems they have not encountered before. This means they should be able to identify the relevant physical principles, make and assess assumptions, choose appropriate methods and evaluate whether their result is meaningful. This is the central view that guides my teaching across all levels, from first-year mechanics to advanced many-body quantum mechanics and student supervision.

### 2.1 Educating problem solvers through student-driven reasoning (*Focus on student learning and my development over time*)

Despite this view being with me since the I began teaching, the *way* I am teaching has developed substantially over time. Initially, I prepared primarily by solving the weekly problem set in advance. When students asked for help, I would often sit down with them and explain the solution step by step. Although this appeared helpful in the moment, I gradually noticed that students often returned with similar questions in later weeks, suggesting that they had followed the solution without developing an independent way of approaching similar problems. I also realised that, by beginning with my own explanation, I learned too little about what the student already understood and did not properly locate the actual misunderstanding.

These experiences led me to an approach I think of as *student-driven reasoning*. When a student now asks for help, I begin instead by asking them to reconstruct their own reasoning: *What have you tried? Why did you choose this approach? What assumptions have you made?* This lets me identify both their current level of understanding and the

conceptual step where they need support. I then use targeted follow-up questions that are challenging but attainable, to help them move their reasoning forward rather than providing the next step directly. This approach has proven to be particularly valuable when teaching groups with diverse educational backgrounds and preparation, such as mixed groups of “Physics and Astronomy” and “Materials Science for Energy and Nanotechnology” students in mechanics at UiO, or mixed-ability supervision groups at Cambridge. By working from students’ own ideas, such an approach gives them a larger sense ownership and mastery of the problem-solving process. I have found that this supports motivation, deeper understanding, and more durable problem-solving skills in problem classes and lectures.

My experience as supervisor of an MSc student working with classical Monte Carlo simulations gave me an opportunity to apply the above principle also in a supervision setting. Through weekly meetings, I helped the student with the ideas behind the numerical implementation and to connect with the physical questions of the project. For example, we discussed possible equilibration schemes and their trade-offs, while the student selected and justified the method ultimately used. While an MSc project needs a clear scope and sufficient progress towards completion, this experience showed me that students can learn more from pursuing and reassessing their own ideas than from simply following the supervisor’s preferred route. My role as a supervisor is therefore to explain the reasoning behind my suggestions, discuss their advantages and limitations, and help the student make informed choices, rather than to decide every step for them. The degree of freedom of course depends on the stage of the project and the particular student. Furthermore, in PhD supervision, the scope for defining and pursuing independent directions should gradually increase with time. As I gain more supervisory experience, I will continue to develop how I balance structure, feedback and student autonomy at different stages of a project and at different student levels.

Student-driven reasoning also goes hand in hand with peer learning, something I now try to use actively as part of my teaching. Early in my group session teaching, I often worked one-to-one with individual students while the rest of the group remained passive. Through the MN Learning Assistant course in 2019 and my subsequent teaching experience, I became more aware that active learning depends not only on asking students the right questions, but also on how the learning environment is organised. I have learned to organise both the physical space and the activities in ways that instead encourage students to engage with each other as well as with me. This includes arranging tables in small groups, sit down with groups rather than remaining physically separate as the teacher and encourage students to discuss ideas. These changes may appear small, but they altered the interaction in the room. Students became more willing to discuss with each other and to involve me in their reasoning process.

My experience is that peer discussions often have high learning potential for all involved parts. Explaining a concept requires students to organise, articulate and thereby deepen their own understanding. At the same time, fellow students often have a particularly good intuition for which aspects of an explanation their peers find difficult. My role is to facilitate and guide the discussions, ensuring that they are productive and inclusive for students with different levels of prior experience. These interactions also help foster a sense of shared engagement within the student group, which I consider important for a positive learning environment.

I see lectures and problem classes as complementary parts of one learning process where lectures should provide direction and a shared conceptual framework, while problem classes should allow students to process ideas, reason through problems, and apply the tools introduced in lectures. Thus, when contributing to a course, I aim to help students see a clear conceptual thread rather than a sequence of unrelated topics. They should understand the overall goal of the course, what they are expected to know already, which new problems are being addressed, and how the tools introduced in one part of the course can be used later. In the lectures I teach, I therefore aim to provide a clear internal structure by discussing why a question matters, which limitations of earlier approaches motivate a new method, which assumptions are being made, and how to connect to other parts of the curriculum.

In the advanced course on many-body quantum physics, which I am currently co-lecturing, this means building new methods from concepts students have already encountered. For example, when lecturing a DMRG session I will begin with the limitations of exact diagonalisation and use them to motivate the search for a compressed representation of a quantum state. Starting from singular-value decomposition, I develop the central idea behind DMRG before moving to high-level pseudocode and a short notebook exercise. Students could then investigate how bond dimension affects accuracy, convergence and computational cost, and benchmark results against a known limit. This lets them encounter DMRG as a response to a physical and computational problem rather than as an isolated algorithm to memorise. The same algorithm also form a natural bridge to quantum simulation later in the course. I would use models such as Hubbard and Heisenberg systems to compare quantum algorithms and hardware with

DMRG and exact diagonalisation. This allows students to examine both the potential and the current limitations of quantum technology, not only how a quantum method works, but what it must outperform, how it can be benchmarked, and what would count as a meaningful quantum advantage.

My focus on student-driven reasoning and peer discussion also applies to lectures. I use planned pauses for students to articulate and test their understanding with peers. In smaller courses, this can take the form of plenary discussion, while in larger cohorts, I envision short peer discussions combined with plenary or anonymous responses can make common difficulties visible. This would give students a break to catch up, helps them process demanding arguments actively and allows me to adapt explanations, examples and pace. I am continuing to develop how I use these activities in different teaching settings, particularly how to balance active participation with the need to provide sufficient structure and pace in larger cohorts.

Inclusion is closely connected to my teaching approach. I am conscious that not all students are equally ready to speak in plenary. I have myself been the quiet student who does not dare to raise a hand, and this informs how I work in both small-group and lecture settings. In problem classes and lectures with smaller audiences, I make a point of talking to all students over the course of the session. When I notice students who are silent or withdrawn, I approach them individually, often beginning with a low-threshold question, such as how they found the previous assignment, before moving to the actual physics. This combination of direct contact, small-group discussions and low-threshold ways of answering helps create a room where it is acceptable to be uncertain and to ask questions, which I would like students to realise is a prerequisite for learning.

I also want students to engage actively with physics in several complementary ways: through reasoning in text and speech, discussion, experiment, analytical exercises and computational exploration. Even as a theoretical and computational physicist, I see considerable value in connecting theory to hands-on experimentation. A particularly clear example of this arose in my mechanics teaching, where many students struggled with the concept of gyroscopes. I found that formal explanations and repeated work on the problem set did not sufficiently develop intuition for the phenomenon. The next time I taught the course, I brought a physical gyroscope to class and allowed students to handle and explore it on their own before we analysed its behaviour together. I then gave a short interactive mini-lecture structured around the conceptual steps I expected to be difficult. This experience reinforced my view that abstract reasoning benefits from concrete anchors, and that teaching should be adjusted when existing explanations do not lead to the intended learning. In quantum courses it is less straightforward to demonstrate phenomena directly, but here I think collaboration with experimental colleagues and coordination with laboratory courses is key, for example by timing discussions of Zeeman splitting or diffraction with the corresponding laboratory exercises, or by using real experimental data as input to numerical analysis.

Computational exploration is another important way in which students can engage actively with physics. My aim in teaching numerical methods is not that students merely learn to run existing code or producing syntactically correct code, but that they understand the principles behind the algorithms, the approximations they rely on, and the regimes in which they fail. They should be able to connect a numerical result to the underlying physical model, test it against known limits, and assess whether it is meaningful. This is particularly important in advanced quantum physics, where numerical methods are often indispensable for problems beyond analytical treatment. When teaching computational methods, such as in FYS4480, I therefore aim to combine theoretical discussion with computational experimentation, allowing students to explore how assumptions, parameters and algorithmic choices affect both numerical accuracy and physical interpretation.

## **2.2 Evidence-informed development of teaching: From course to study-programme level** (*A researching approach*)

My development as a teacher has also involved becoming more aware about collecting different forms of feedback during teaching. I regularly use short quizzes, giving students a low-threshold opportunity to test their understanding, while providing me with immediate information about concepts that require further attention. Mid-term evaluations, submitted work and classroom discussions can similarly reveal where students encounter difficulties, whether an explanation should be approached differently, or whether a different learning activity would be more productive. Over time, these sources can help identify recurring difficulties and indicate whether adjustments to teaching support student learning.

Assessment provides an important source of evidence about whether students have achieved the intended learning outcomes. For this evidence to be useful, the assessment itself must be designed and evaluated carefully. In 2020, I carried out a project on the validity of multiple-choice questions in physics examinations at UiO. I analysed student performance in three examinations containing both multiple-choice and conventional long-answer questions and found moderate to high correlations between performance on the two formats. Questions requiring calculation or otherwise discriminating well between different levels of performance contributed most positively to this relationship. The project showed that carefully designed multiple-choice questions can provide a comparable indication of student achievement to conventional long answer questions, and also thought me to treat assessment design as an educational question that can be investigated systematically.

I furthermore see teaching as a process extending beyond the individual course. To ensure student progression, it is important to strive for a coherent programme where courses build on and reinforce each other. A formative experience in this regard was the Glittum–Haugerud report, which I co-authored as a student evaluation of the Physics and Astronomy bachelor's programme at UiO in 2019. Combining our own experiences with a survey of 52 current and former students, the report examined the study programme from individual learning activities to its overall structure. It addressed, among other issues, students' sense of belonging, attendance and learning activities in problem classes, compulsory coursework, experimental and numerical physics and conceptual understanding. It also presented two detailed proposals for a broader restructuring of the programme. The report has been an important input to subsequent discussions of the bachelor's programme and its development.

### **2.3 Educational quality as a collective academic responsibility (*A collegial attitude and practice*)**

I view educational quality as a collective academic responsibility. Teaching development benefits when questions about student learning, assessment and course design are made visible and discussed openly among colleagues. Both my multiple-choice assessment project and the Glittum–Haugerud report was presented at the Department of Physics' biannual seminar, making the educational questions addressed visible and open to collective discussion.

Collegial dialogue is particularly important for coherence at study-programme level. At the bachelor's level, where many students take similar courses, it would be beneficial if courses build explicitly on one another and reinforce shared concepts and skills. Achieving this requires lecturers and teachers to be familiar not only with formal course descriptions, but also with how preceding and parallel courses are taught in practice. It therefore depends on regular dialogue between colleagues. For example, dialogue between those teaching FYS1400, FYS2140 and FYS3110 can help make the progression in quantum mechanics more explicit for students. Similarly, programming skills introduced in IN1900 can be used in parallel in physics courses such as FYS1100. This both strengthens students' programming skills and makes the relevance of mathematics and computer science more visible in their physics education.

Through the NOKUT evaluation, the Glittum–Haugerud report and subsequent committee work, I have contributed to discussions of curriculum structure, student participation, progression and the relationship between education and working life. In the QST programmes, I would bring this collegial perspective to course and programme development, supporting coherent progression across theoretical, experimental and computational components.

### **2.4 Teaching in the age of AI**

Generative AI changes the conditions under which students work with problem solving, conceptual understanding and assignments. It can provide explanations, code and plausible-looking solutions in seconds, and can be a useful resource when used responsibly, for example to obtain an alternative explanation. At the same time, easy access to apparently complete solutions can turn an assignment into an exercise in following an answer. This challenge is closely related to my teaching philosophy. Students develop deeper and more durable understanding when they are encouraged to reason through a problem themselves, rather than being given a complete solution too early. I see it as our responsibility as teachers to rethink how we teach to facilitate such learning in the age of AI.

This does not mean that learning should take place only in group settings. Individual effort is essential, as much of the learning takes place when students struggle with a problem, test possible approaches, and identify what they do and do not understand. Group sessions, peer discussion, and on-campus activities are most valuable when they build on such prior work. They can then provide opportunities for students to articulate their reasoning, compare approaches, ask questions, and receive feedback.

The challenge in the age of AI is therefore not simply to discourage students from using AI, but to design learning activities that make independent preparation worthwhile. This may involve tasks that require students to explain their reasoning, document intermediate steps, discuss unsuccessful approaches, or apply ideas to a new problem. If group sessions are designed to build on this preparation and facilitate further discussion, students can feel a deeper sense of mastery and hopefully realise that this way of learning is more valuable long term. This raises of course further challenges, like how to make students show up and engage in the first place? I believe part of the answer lies not only in how sessions are designed, but whether students feel they belong in the learning environment.

## **2.5 Belonging as a foundation for recruitment and retention**

Belonging develops through repeated experiences of contributing in a safe and inclusive environment, discussing ideas with peers, and being recognised as part of a community. Once students feel this sense of belonging, they are in turn more likely to show up, participate, and take ownership of their own learning and that of their peers. This again reinforces the sense of belonging further, and also illustrates how a sense of belonging is important for retention.

Belonging also matters for recruitment. Visible and inclusive academic communities shape who is able to imagine themselves belonging in physics and QST. To educate the next generation of physicists and quantum technologists, we must attract students from the broadest possible pool and ensure that they encounter an inclusive environment once they arrive. I have long regarded outreach as part of educational work. My outreach activities have ranged from school visits that introduce pupils to physics and STEM studies to welcome-week and career activities that help students already enrolled see possible paths ahead and communication through social media. As one of relatively few women at my career stage in theoretical physics, I also see visibility and role modelling as part of my responsibility. Who stands at the front of a classroom, a school visit, or a public event can influence who is able to imagine themselves belonging in physics later, and who feels they belong during their studies.

I would like to contribute to recruitment and retention in the QST and Physics study programmes through outreach, undergraduate teaching, and keep arranging for low-threshold academic and social meeting places for students. For example, I have started organising bi-weekly meetings for bachelor's students (currently 20 students participating) to get experience with combining numerics and quantum mechanics. This is not only with an aim of educating the students, but also to give them a separate arena to both socialise with each other in an academic setting, get to work closer with academics and see themselves as part of an academic community early in their studies.

## **2.6 Outlook**

To summarise, I have moved from seeing teaching primarily as the delivery of prepared explanations to seeing it as the design of learning environments in which students can reveal, test and develop their own understanding, with a goal of educating independent problem solvers with a deep physical understanding. In student supervision, lectures and problem classes, I therefore emphasise a clear conceptual thread, active learning, guided problem solving, peer discussions and opportunities for students to articulate and test their reasoning. At the same time, I seek to create an inclusive learning environment in which students feel safe to participate and develop confidence.

I also understand educational quality as a collective responsibility extending beyond the individual course. Through programme development, committee work and collaboration with colleagues, I have contributed to discussions of progression, the integration of theoretical, experimental and computational competence and students' academic and social belonging. As AI changes how students access explanations, solutions and code, I will place even greater emphasis on independent judgement, transparent reasoning, peer interaction and the ability to validate computational results.

In the coming years, I aim to develop further as a lecturer, course designer and supervisor. Building on my experience from problem classes, smaller-group teaching, lecturing and supervision, I want to work on better incorporating my experiences and vision into the lecture and course design arena. Further, I want to continue developing how I balance structure, active learning and student autonomy in different teaching settings. I plan to complete the required formal pedagogical training course, and to use student feedback and collegial dialogue more systematically, to evaluate and improve my teaching over time. Overall, I have a strong wish to contribute to the continued development of coherent, inclusive and research-connected physics and QST programmes at UiO.

## References – Cecilie Glittum

1. Claudio Castelnovo  
Relation: Collaborator and PhD supervisor  
E-mail: cc726@cam.ac.uk  
Phone: +44(0)1223 3 37433
2. Olav Fredrik Syljuåsen  
Relation: Collaborator and MSc supervisor  
E-mail: sylju@uio.no  
Phone: +47 411 78 005
3. Morten Hjorth-Jensen  
Relation: Collaborator and co-lecturer  
E-mail: mhjensen@uio.no  
Phone: +47 482 57 387



# Cecilie Glittum

*Curriculum Vitae*

## PERSONAL DETAILS

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|--------------------|--------------------------------------------------------------------|
| <i>Born</i>        | 11. November, 1996                                                 |
| <i>Nationality</i> | Norwegian                                                          |
| <i>Address</i>     | Uelands gate 61H, 0460 Oslo, Norway                                |
| <i>Phone</i>       | +47 417 68 728                                                     |
| <i>E-mail</i>      | <a href="mailto:cecilie-gl@hotmail.com">cecilie-gl@hotmail.com</a> |

## POSITIONS

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### MSCA Postdoctoral Research Fellow

2025-

*University of Oslo*

Project leader of the DSPINN project, where we investigate the possibilities of stabilising quantum spin liquids from kinetic frustration.

### Postdoctoral Researcher

2024-2025

*Helmholtz-Zentrum Berlin für Materialien und Energie*

Numerical study of topological excitations in spiral spin liquids.  
Supervising a master's student.

## EDUCATION

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### PhD in Physics

2020-2024

*Trinity College, University of Cambridge*

Theory of Condensed Matter group at the Cavendish Laboratory.

Thesis title: *Wonderland of frustration: Magnetic moments and itinerant electrons on the pyrochlore lattice.*

Date of defence: 25.10.2024

### MSc. in Theoretical Physics

2018-2020

*University of Oslo*

Thesis title: *Nematic Bond Theory of Frustrated Heisenberg Models on Triangular and Honeycomb Lattices.*

GPA: 5.00 (A)

### BSc. in Physics, Astronomy and Meteorology

2015-2018

*University of Oslo*

Specialisation in physics.

Research project title: *Field Theory of Frustrated Magnets.*

GPA: 4.94 (A)

## TEACHING EXPERIENCE

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### Lecturing

2025-2026

*University of Oslo*

- FYS4480 Quantum mechanics for many-particle systems: Co-responsible for organising and lecturing (2026)
- FYS4110 Advanced quantum mechanics: Guest lecture on quantum magnetism (2025)

### Teaching assistant

2017-2023

*University of Cambridge*

Supervising third year undergraduate students in Part II Advanced Quantum Physics (2020), giving example classes in Theoretical Physics I (classical field theory) (2020, 2021, 2022, 2023) and Part II Computational Physics (2023).

*University of Oslo*

Teaching assistant in FYS-MEK 1110 - *Mechanics* (2017, 2019) and MAT-INF 1100 - *Modelling and Computations* (2017). As a teaching assistant, I got pedagogical training for six months through weekly seminars (learning assistant programme, spring 2019).

### Course development

2018-2023

*University of Cambridge*

Constructed the problem set on neural networks for Part II Computational Physics.

*Simula Research Laboratory/University of Oslo*

Constructed the problem sets used in two courses given at the University of Oslo: IN1900 - *Introduction to Programming with Scientific Applications* and IN1910 - *Programming with Scientific Applications*. The problems were on a wide range of topics from introductions to the programming languages to more advanced problems on object orientation, both in Python and C++.

### Evaluation of multiple-choice questions in physics exams

2020

*University of Oslo*

Statistical study of correlations between answers to multiple-choice questions and conventional free-response questions in physics exams. The job involved writing of a report and giving a presentation at the Department's seminar on education.

### Writing Science instructor

2019

*University of Oslo*

Teaching a course on academic writing for first year master's students at the Department of Physics (2019).

### Study programme development

2019

*University of Oslo*

Suggesting changes to the structuring and curriculum of the bachelor's programme in Physics and Astronomy at the Department of Physics, University of Oslo. The job consisted of writing a report and giving a presentation at the Department's seminar on education.

## AWARDS AND FUNDING

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- Marie Skłodowska-Curie Actions Postdoctoral Fellowship (COFUND) 2025-2028
- Institute of Physics Sam Edwards PhD Thesis Prize: Theory of Condensed Matter Group 2025, runner-up
- **American Physical Society Outstanding Dissertation in Magnetism Award 2025**
- Martin Landrøs pris for fremragende mastergrad i fysikk 2021 (EN: Martin Landrø's Prize for Outstanding Master Thesis in Physics 2021) awarded by the Norwegian Physical Society
- Aker Scholarship 2020 (4 years of PhD studies in Cambridge fully funded)

## PROPOSALS UNDER EVALUATION

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- *To be submitted September 2026:* Nobel Symposium 2028: Quantum Spin Liquids, led by Maria Hermanns
- *Submitted June 2026:* QuNorth Flagship, EIGEN-Q, led by Morten Hjorth-Jensen

## OTHER ACADEMIC EXPERIENCE

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### Committees

#### *University of Oslo*

- Fixed-term representative on the Faculty Board for The Faculty of Mathematics and Natural Sciences (2026)
- Norwegian Agency for Quality Assurance in Education (NOKUT) evaluation of labour-market relevance in disciplinary programmes (BSc in Physics and Astronomy at the University of Oslo) (2019 - 2020)
- Programme Council of the Bachelor's Programme in Physics and Astronomy (2019 - 2020)
- Programme Council of the Master's Programme in Physics (2018 - 2020)
- The Board of the Department of Physics (deputy) (2017 - 2019)

#### *University of Cambridge*

- Secretary of Trinity College BA Society (2021-2022)
- Student representative in the Liaison Committee, Trinity College (2021-2022)

### Referee

APS Physical Review: Letters (1), B (7), Research (4)

## Selected invited talks

- **Scheduled: *Kinetic frustration as a new route to quantum matter*, The 14th International Conference on Highly Frustrated Magnetism HFM27, Munich, Germany (July 2027)**
- Scheduled: *Kinetic frustration as a new route to quantum matter*, dScience Lunch Seminar, University of Oslo, Norway (Spring 2027)
- *RVB physics from kinetic magnetism*, Advanced Working Group on Kinetic Magnetism, Cambridge, UK (2026)
- *Doping-induced quantum spin liquids: Towards robust topological order for quantum technology*, IQT Nordics Junior Researcher Symposium, Oslo (2026)
- *A resonant valence bond spin liquid in the dilute limit of doped frustrated Mott insulators*, Nordic Conference on Recent Trends in Condensed Matter Physics, Niels Bohr Institute, Copenhagen (2025)
- *Doping-induced RVB spin liquid in a frustrated Mott insulator*, Theory Seminar, University of Oslo (2025)
- *Field-induced Multic $\vec{Q}$  States in a Pyrochlore Heisenberg Magnet*, TCM seminar, Cavendish Laboratory, University of Cambridge (2025)
- ***Wonderland of frustration: Magnetic moments and itinerant electrons on the pyrochlore lattice*, Award talk at APS Global Physics Summit (March Meeting), Anaheim (2025)**
- *Doping-induced spin liquid in a pyrochlore magnet*, Smalley-Curl Institute, Rice University (2024)
- *Doping-induced spin liquid in a pyrochlore magnet*, Los Alamos National Laboratory, Los Alamos (2024)
- *Sublattice pairing in pyrochlore Heisenberg antiferromagnets*, KTH Stockholm (2024)
- *Sublattice pairing in pyrochlore Heisenberg antiferromagnets*, Helmholtz-Zentrum Berlin (2023)
- *Sublattice pairing in pyrochlore Heisenberg antiferromagnets*, Theory Seminar, University of Oslo (2023)

## Outreach

- Welcome-week seminar for bachelor's students in quantum technology, University of Oslo (2026)  
Title: *Når magneter blir frustrerte: Fra kjøleskapsmagneter til kvantespinnvæsker.*
- Fysikk i Origo (seminar series arranged by the Student Physics Association), University of Oslo (2026)  
Title: *Frustrated magnets and quantum spin liquids.*
- Seminar on academic career for final year master's students in physics, University of Oslo (2026)  
Title: *The Life of a Postdoc.*
- Welcome week seminar for new bachelor's and master's students in physics, University of Oslo (2026)  
Title: *Academic career: How to prepare during your studies.*
- Prepare Project (2016-2019), Simula School of Research and Innovation  
Visiting high schools in the Oslo-region giving presentations about natural sciences and study opportunities, with an aim to motivate and inspire towards studies in STEM.

## SKILLS

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|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Languages</i>   | Norwegian (Mother tongue), English (Fluent), German (Basic)                                                                                |
| <i>Programming</i> | C++, PYTHON, MATHEMATICA                                                                                                                   |
| <i>Techniques</i>  | Exact diagonalisation, tensor networks/matrix product states,<br>Nematic Bond Theory (field-theoretic framework),<br>classical Monte Carlo |

# Publication List - Cecilie Glittum

## Publications:

- (1) **Cecilie Glittum**, Han Yan and Johannes Reuther, “Fractional vortices in a spin-isotropic spiral spin liquid”, [arXiv:XXX](#), (2026) [submitted to Nat. Commun.].
- (2) **Cecilie Glittum** and Olav F. Syljuåsen, “Weak first-order phase transition out of the classical kagome spin liquid”, [arXiv:2605.26779](#), (2026) [submitted to Phys. Rev. Lett.].
- (3) Fanjun Xu, Ralf Feyerherm, **Cecilie Glittum**, Thomas J. Hicken, Hubertus Luetkens, Jonas A. Krieger, Cintli Aguilar-Maldonado, Sven Luther, Lucy K. Saunders, Clemens Ritter, Peter Fouquet, Margarita Russina, Karel Prokeš, A.T.M. Nazmul Islam and Bella Lake, “Dynamical magnetism in the disordered cubic lattice material  $\gamma$ -Ba<sub>3</sub>CoNb<sub>2</sub>O<sub>9</sub>”, [arXiv:2604.18794](#), (2026) [submitted to Phys. Rev. B].
- (4) Zhao Zhang and **Cecilie Glittum**, “Resonating Valence Bond Ground States on Corner-sharing Simplices”, [arXiv:2507.10471](#), [SciPost Phys. \*\*20\*\*, 121 \(2026\)](#).
- (5) **Cecilie Glittum** and Olav F. Syljuåsen, “The Finite-Temperature Behavior of a Triangular Heisenberg Antiferromagnet”, [arXiv:2510.02042](#), [J. Phys.: Condens. Matter \*\*38\*\*, 135801 \(2026\)](#).
- (6) **Cecilie Glittum** and Olav F. Syljuåsen, “Field-induced Multi- $\vec{Q}$  States in a Pyrochlore Heisenberg Magnet”, [arXiv:2503.05353](#), [Phys. Rev. B \*\*111\*\*, 214444 \(2025\)](#).
- (7) **Cecilie Glittum**, Antonio Štrkalj, Dharmalingam Prabhakaran, Paul A. Goddard, Cristian D. Batista, and Claudio Castelnovo, “A resonant valence bond spin liquid in the dilute limit of doped frustrated Mott insulators”, [arXiv:2408.03372](#), [Nature Physics \*\*21\*\*, 1211–1216 \(2025\)](#).
- (8) **Cecilie Glittum** and Olav F. Syljuåsen, “Sublattice pairing in pyrochlore Heisenberg antiferromagnets”, [arXiv:2305.09409](#), [Phys. Rev. B \*\*108\*\*, 014413 \(2023\)](#).
- (9) **Cecilie Glittum**, Antonio Štrkalj, and Claudio Castelnovo, “Emergence of quasiperiodic behavior in transport and hybridization properties of clean lattice systems”, [arXiv:2210.02480](#), [Phys. Rev. Research \*\*5\*\*, 013083 \(2023\)](#).
- (10) Giuseppe De Tomasi, Oliver Hart, **Cecilie Glittum**, and Claudio Castelnovo, “Effects of critical correlations on quantum percolation in two dimensions” [arXiv:2203.14977](#), [Phys. Rev. B \*\*105\*\*, 224316 \(2022\)](#).
- (11) **Cecilie Glittum** and Olav F. Syljuåsen, “Arc-shaped structure factor in the  $J_1$ - $J_2$ - $J_3$  classical Heisenberg model on the triangular lattice”, [arXiv:2108.09120](#), [Phys. Rev. B \*\*104\*\*, 184427 \(2021\)](#).

## Dissertations:

- (12) PhD thesis: “Wonderland of frustration: Magnetic moments and itinerant electrons on the pyrochlore lattice”, (2024) [<https://doi.org/10.17863/CAM.114778>].
- (13) Master thesis: “Nematic Bond Theory of Frustrated Heisenberg Models on Triangular and Honeycomb Lattices”, (2020) [<http://urn.nb.no/URN:NBN:no-81891>].